

Algorithmic Pseudocode Sample 13

Source-backed Image2Struct algorithm sample

1 Algorithm

This sample contains algorithmic pseudocode extracted from a source-backed LaTeX benchmark dataset. It is wrapped in a minimal article document for pdfLaTeX validation.

Algorithm 1 Source-backed algorithmic procedure

- 1: Initialize actor network μ_θ and critic networks Q_{ω_1} and Q_{ω_2} .
- 2: Initialize corresponding target networks: $\theta' \leftarrow \theta$, $\omega'_1 \leftarrow \omega_1$, $\omega'_2 \leftarrow \omega_2$ and choose $\rho_p \in (0, 1)$.
- 3: Initialize replay buffer $\mathcal{R}$.
- 4: Choose a batch size K , a gradient based optimization algorithm and a corresponding learning rate $\lambda_{\text{actor}}, \lambda_{\text{critic}} > 0$ for both optimization procedures, a time step size Δt and a stopping criterion.
- 5: Choose standard deviation exploration noise σ_{expl} and lower and upper action bounds $a_{\text{low}}, a_{\text{high}}$.
- 6: **repeat**
- 7: Select clipped action and step the environment dynamics forward.

$$a = \text{clip}(\mu_\theta(s) + \epsilon, a_{\text{low}}, a_{\text{high}}), \quad \epsilon \sim \mathcal{N}(0, \sigma_{\text{expl}}).$$

- 8: Observe next state s' , reward r , and done signal d and store the tuple (s, a, r, s', d) in the replay buffer.
- 9: **if** s' is terminal **then**
- 10: Reset trajectory.
- 11: **end if**
- 12: **for** j in range(*update frequency*) **do**
- 13: Sample batch $\mathcal{B} = \{(s^{(k)}, a^{(k)}, r^{(k)}, s'^{(k)}, d^{(k)})\}_{k=1}^K$ from replay buffer.
- 14: Compute targets (Clipped Double Q-learning and policy smoothing).

$$y(r, s', d) = r + (1 - d) \min_{i=1,2} \{Q_{\omega'_i}(s', \tilde{a})\}, \quad \tilde{a} = \text{clip}(\mu_{\theta'}(s) + \epsilon, a_{\text{low}}, a_{\text{high}}), \quad \epsilon \sim \mathcal{N}(0, \sigma_{\text{target}}).$$

- 15: Estimate critic gradient $\nabla_{\omega} L(Q_{\omega}^{\mu_\theta})$ by

$$\nabla_{\omega_i} \left(\frac{1}{K} \sum_{k=1}^K \left(Q_{\omega_i}(s^{(k)}, a^{(k)}) - y(r^{(k)}, s'^{(k)}, d^{(k)}) \right)^2 \right), \quad \text{for } i = 1, 2.$$

- 16: Update the critic parameters ω_i based on the optimization algorithm.
- 17: **if** $j \bmod \text{policy delay frequency} = 0$ **then**
- 18: Estimate actor gradient $\nabla_{\theta} J(\mu_\theta)$ by

$$\nabla_{\theta} \left(\frac{1}{K} \sum_{k=1}^K Q_{\omega_1}(s^{(k)}, \mu_\theta(s^{(k)})) \right).$$

- 19: Update the actor parameters θ based on the optimization algorithm.
- 20: Update target networks softly:

$$\theta' \leftarrow \rho_p \theta' + (1 - \rho_p) \theta, \quad \omega'_i \leftarrow \rho_p \omega'_i + (1 - \rho_p) \omega_i, \quad \text{for } i = 1, 2.$$

- 21: **end if**
 - 22: **end for**
 - 23: **until** stopping criterion is fulfilled.
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